Lecture plan · for the lecturer

Real-time and the power budget

IoT · University · Lecture 2

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Budget energy with duty cycles.
- Define hard and soft real-time.
- Check schedulability.

Activity: Measure, then schedule

1. Warm-up (5 min): Show the nine-day node's current trace and ask students to find the problem.
2. Measure (20 min): Teams log active and sleep current on their node.
3. Budget (15 min): Compute I_avg and predicted life, then halve the wake time and recompute.
4. Schedule (15 min): Compute U for three tasks and test the rate-monotonic bound.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- How far was the datasheet from your measurement?
- Is your system hard or soft real-time?
- What would you send less often?

Exit ticket

- Compute the battery life of a node at 0.25 mA average on 1,000 mAh.

Try it in your next lecture
Give each team a current meter and its own node; they measure active and sleep current, predict battery life and defend the number.

Seminar sheet · for students

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Name _____ Group _____ Date _____

1 **Tasks (C = 1 ms, T = 4 ms) and (C = 2 ms, T = 10 ms). Under the 2-task RM bound of about 83%?**

◯ No, U = 120% ◯ Yes, U = 30% ◯ Yes, U = 45%

2 **Work out the battery life.**

Active 30 mA for 2 s every 60 s, sleep ≈ 0. I_avg = _____ mA, so 2,000 mAh lasts _____ hours.

Word bank: 30 · 1 · 2000 · 60

3 **Which changes extend battery life most for a radio sensor? Choose all that apply.**

☐ Shorten time-on-air by sending fewer bytes ☐ Send a daily summary instead of every reading

☐ Use the deepest sleep mode between readings ☐ Run the processor at full speed all the time

4 **A real-time system is one that runs as fast as possible.**

◯ True ◯ False

5 **Which of these tasks are hard real-time? Choose all that apply.**

☐ Weather dashboard refresh ☐ Video frame display ☐ Airbag trigger ☐ Pacemaker pulse

Answers · for the lecturer

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1 **Yes, U = 45%**

U = 1/4 + 2/10 = 0.45, well under 0.83.

2 **Active 30 mA for 2 s every 60 s, sleep ≈ 0. I_avg = 1 mA, so 2,000 mAh lasts 2000 hours.**

D = 2/60, so 30 × 2/60 = 1 mA, and 2,000 mAh / 1 mA = 2,000 h.

3 **Send a daily summary instead of every reading; Use the deepest sleep mode between readings; Shorten time-on-air by sending fewer bytes**

Radio time and sleep current dominate; a processor always at full speed wastes energy.

4 **False**

Real-time means predictable: every task meets its deadline.

5 **Airbag trigger; Pacemaker pulse**

Ask: is a late answer a wrong answer? If so, it is hard real-time.

For you to think about

How do your students currently estimate power? What would change if they had to measure it?